A Theory of Economic Slack

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CHAPTER 3.

Research on which this book is based

This book is based for the most part on research that I conducted during graduate school at the University of California–Berkeley under the supervision of George Akerlof and Yuriy Gorodnichenko, and after it in collaboration with Emmanuel Saez.

This chapter briefly summarizes that research. It also explains what motivated the research, and provides the key references on which the book is based. The purpose of the chapter is not to describe results, but to explain how the theory presented in the book emerged from a sequence of concrete research questions, themselves motivated by interconnected macroeconomic puzzles and real-world policies.

3.1. Explaining queues of job seekers in recessions

Early on in graduate school I read about Okun's misery index—the sum of the unemployment rate and inflation rate, designed as a simple measure of welfare in the US economy.[1] A low index means that the economy is doing well; a high index that it is doing poorly. The misery index posits that inflation and unemployment are the two main diseases affecting developed economies. The discovery of the misery index prompted me to start working on inflation. I particularly wanted to understand why seemingly tiny changes in the price of a baguette in France triggered such outrage from customers. But that project proved to be too challenging, so I turned to the other component of the misery index: unemployment.

A first book that George advised me to read to start thinking about unemployment and

[1] See Asher, Defina, and Thanawala (1993) for a history of the misery index.

business cycles was *Prices & Quantities* by Arthur Okun (1981). In it, Okun sketches a new macroeconomic framework in which labor and product markets are organized around implicit contracts between sellers and buyers. Although the slackish business cycle model developed in the book is quite different from the framework that Okun had in mind, the book draws in numerous places on evidence that Okun collected, observations that he made, and references that he provided. Generally, the book shares Okun's conviction that Walrasian theory is inappropriate to describe labor and product markets, and that prices and wages are determined by norms that are somewhat insulated from cyclical forces.

George also suggested that I read *Equilibrium Unemployment Theory* by Christopher Pissarides (2000)—the leading textbook on unemployment theory. The textbook introduces the Diamond-Mortensen-Pissarides (DMP) model of the labor market, which was developed by Pissarides together with Peter Diamond and Dale Mortensen in the 1970s and 1980s. For their work, the three of them received the Nobel Prize in Economics in 2010.[2]

Reading the textbook, it struck me that the DMP model does not allow for the possibility—which seemed relevant in real life—that sometimes the economy just does not have enough jobs for all workers. This is an old idea in macroeconomics—a very Keynesian idea. Keynes (1936) believed that in bad times there are just not enough jobs for everyone at the going wage: jobs are rationed. The literature on efficiency wages built models producing job rationing, mostly in the 1980s, but they fell out of favor, maybe because they could not easily be integrated in the fast-growing DMP framework.[3]

By ignoring job rationing, the DMP model cannot provide a good description of what happens in the labor market. In bad times, in the real world, workers queue for jobs. For instance, traditional pictures from the Great Depression show workers lined up in front of factory gates or at job bureaus; the same queues appeared in Charlie Chaplin's *Modern Times*. In that situation, a factory manager can just pick any worker that they want immediately; it is costless for firms to hire; matching frictions are a non-issue. And yet, workers visibly queue for jobs. So it seems that something else happens in this situation: jobs are lacking.

Thus, I became convinced that to describe the labor market better, we must allow for job rationing. My PhD thesis therefore developed a labor market model that features not only frictional unemployment—as in the DMP model—but also rationing unemployment—as in the *General Theory*. In the model, firms might not want to create enough jobs for all workers who want to work. The main result of the thesis is that in recessions, frictional

[2] Albrecht (2011) offers a wonderful overview of the early work by Diamond, Mortensen, and Pissarides, and of the literature that built on it. Diamond (2011), Mortensen (2011), and Pissarides (2011) provide their own perspective on their research and the literature in their Nobel lectures.

[3] For surveys of the efficiency-wage literature, see Akerlof (1984), Yellen (1984), and Katz (1986). See Akerlof and Yellen (1990) for one of the most mature models in that literature.

unemployment is small and most of unemployment is accounted for by job rationing. This core idea was then published in the *American Economic Review* in 2012 (Michaillat 2012).

The AER paper introduces one of the book's key ideas: that a matching structure allows us to bridge the gap between older, Keynesian theories that emphasize the role of demand in macroeconomics and modern theories that emphasize the role of supply. In doing so, the book offers a model that accounts for both demand-side and supply-side factors simultaneously, and that can be used to analyze both demand-side and supply-side policies, as well as policies influencing both supply and demand simultaneously. This idea is found throughout the book, but a streamlined version of the labor market model from the AER paper is presented in chapter 11.

A second idea in the paper, which is also an important part of the book, is that matching models are well suited to accommodate realistic, somewhat rigid price and wage norms, and that such price and wage rigidity generates realistic business cycle fluctuations. That idea was developed by Hall (2005), which showed that fixed wages are compatible with bilateral efficiency in matching models, and that with fixed wages labor market matching models generate realistic labor market fluctuations. A further insight from Blanchard and Gali (2010) is that in fact, wages don't have to be fixed: they can simply be a rigid function of productivity. In the real world, very few prices are determined at auction, as the Walrasian model assumes. Instead, most sellers and buyers are engaged in repeated, long-term relationships where norms of fairness matter a great deal. In a paper with Erik Eyster and Kristof Madarasz, that stemmed from the baguette project, but greatly developed and improved it, we document these norms and model pricing under fairness concerns (Eyster, Madarasz, and Michaillat 2021). The book does not use the pricing model developed in that paper, but it uses several of the facts documented there to build realistic price norms, especially in chapters 6 and 7.

Towards the end of my PhD thesis, I stumbled on an interesting property of the model: it is highly state dependent, in that it operates very differently in booms and slumps. As a result, some policies are systematically more effective when the economy is slack, while other policies are systematically more effective when the economy is tight. For instance, the fiscal multiplier is larger when unemployment is high, thus explaining Yuriy and Alan Auerbach's discovery that fiscal multipliers are higher during recessions (Auerbach and Gorodnichenko 2012). This result was published in the *American Economic Journal: Macroeconomics* in 2014 (Michaillat 2014). The state dependence is present throughout the book in various contexts, but especially in chapter 9, where the effects of shocks are strongly slack dependent, and in chapters 12 and 14, where we obtain slack-dependent policy multipliers.

3.2. Should unemployment insurance be extended in recessions?

I went on the job market in 2009–2010, in the aftermath of the Great Recession. One of the big policy questions at the time was whether to make unemployment insurance more generous. Emmanuel attended a mock job talk that I gave in December 2009, thought that the model that I had developed in my thesis could be helpful to think about the question, and reached out to see if I wanted to collaborate with him and Camille Landais, who was doing a postdoc at Berkeley at the time, on a project to determine how unemployment insurance should be adjusted over the business cycle.

The reason why the model was promising is that public finance only took into account supply-side factors—how unemployment insurance affected job search—but did not have the tools to include demand-side factors—the number of workers that firms are willing to hire. The key idea was that if firms do not want to hire more workers, then searching more for jobs will only result in a giant rat race, but not additional employment, which would severely reduce the purported social cost of higher unemployment insurance and make an unemployment insurance extension in bad times more desirable.

So in the summer of 2010, Camille, Emmanuel, and I started working on the following question: how should the generosity of unemployment insurance respond to unemployment fluctuations? We found that in the presence of job rationing, optimal unemployment insurance is more generous than the conventional Baily (1978)-Chetty (2006) level when the labor market is inefficiently slack and below it when the labor market is inefficiently tight. We also show how to identify empirically the presence of job rationing: there is job rationing if and only if the macro effect of unemployment insurance on employment is less than its micro effect. We collect evidence of job rationing in the US labor market. The implication is that unemployment insurance should be more generous in bad times than in good times—as it is in practice in the United States. The project was ultimately published in 2018, in a pair of papers in the *American Economic Journal: Economic Policy* (Landais, Michaillat, and Saez 2018b,a). This analysis of unemployment insurance is discussed in chapter 12.

The AEJ papers aimed to express the formula for optimal unemployment insurance in sufficient statistics. The idea was to describe optimal policy with formulas expressed in terms of statistics that apply across a range of models (not just a specific model) and that are directly measurable (unlike model parameters).[4] Sufficient statistics make the policy tradeoffs transparent and rooted in empirical evidence, but they were not used to study macro, market-level effects of policy. Market-level effects of policies were typically studied in clunky, structural models that kept the policy tradeoffs and mechanisms obscure. The AEJ papers brought sufficient statistics to the macrodesign of policy; the book follows this

[4]See Chetty (2009) and Kleven (2021) for surveys. Emmanuel's job-market paper played a central role in the shift towards sufficient statistics in public econommics (Saez 2001).

methodology, especially in chapter 10 and in part V.

3.3. Connecting aggregate demand to labor demand

Despite the difficulties encountered in our project on unemployment insurance, Emmanuel and I decided to keep working together with the framework. A natural question that arose is why labor demand is sometimes high and sometimes low. In the labor market model from my thesis, which we then used in our unemployment insurance project, fluctuations in labor productivity lead to fluctuations in labor demand. This is just the same as in the DMP model. But in reality, labor productivity is unlikely to be the main driver of business cycles. Surely aggregate demand fluctuations, stemming from changes in monetary policy or animal spirits, are bound to influence labor demand. The question is therefore how to connect aggregate demand to labor demand.

The answer, it turns out, is slack on the product market. Unemployment is the most studied form of slack but slack also exists on the product market: firms are always looking for customers to buy their products and services. We generalize the approach to a model with both the product market and the labor market, using the architecture developed by Barro and Grossman (1971). The resulting model blends three components of unemployment: Keynesian and classical components, in the sense of Malinvaud (1977, figure 3), and a frictional component, as in the DMP model. Unemployment has a Keynesian component because it depends on how difficult it is for firms to sell their goods; it has a classical component because it depends on the real wage; and it has a frictional component because it depends on how costly it is for firms to recruit workers. This work was published in the *Quarterly Journal of Economics* in 2015 (Michaillat and Saez 2015), and is covered in chapter 16.

While our QJE paper offers a simple model that is useful to understand why slack exists and how slack on the different markets interact, it is static so cannot be used to think about interest rates, which inherently involve dynamic savings decisions. And interest rates are the tool used by central banks to stabilize the economy in modern times. So, Emmanuel and I next developed a dynamic model and inserted interest rates into it. In this monetary model, unemployment is determined by the intersection of an aggregate-demand curve, stemming from households' consumption-saving decisions, and an aggregate-supply curve, corresponding to the Beveridge curve. Monetary policy influences the real interest rate, which is a key determinant of the aggregate-demand curve, so monetary policy can be used to stabilize the economy. This work was published in the *Oxford Economic Papers* in 2022 (Michaillat and Saez 2022). The model is covered in chapter 14 and the implications for monetary policy are covered in chapter 17.

3.4. Allowing for long zero-lower-bound episodes

One of the challenges that we faced in developing the OEP paper was how to build a nondegenerate aggregate demand curve. By nondegenerate I mean an aggregate demand curve that is downward-sloping, just like the old IS curve, and that can be used to study a monetary model at the zero lower bound or away from it, without requiring any specific monetary rules. The standard aggregate demand, arising from a standard Euler equation, actually fails these requirements, because it is perfectly elastic: it only tolerates a real interest rate equal to the time discount factor.

We discovered that it was possible to obtain a nondegenerate aggregate demand curve by assuming that people had wealth in their utility function. The justification for the assumption is that wealth is a marker of social status, and people value high social status. Thanks to this assumption, the model featured a nondegenerate aggregate demand curve. In particular, the model behaved well at the zero lower bound, even if the zero-lower-bound episode lasted a very long time, just like in the aftermath of the Great Recession in the United States.

The assumption of wealth in the utility function is extremely useful to produce a well-behaved aggregate demand curve, so it is used throughout part IV and part V. That assumption had been used sporadically in various macroeconomic contexts since Kurz (1968) first introduced it, especially to understand saving and investment behavior, and relatedly, long-run growth.[5] But it had not been used in business cycle research, especially in combination with a slackish model, so the paper, written in 2014, was not published before 2022.

What we did manage to publish was a related paper showing that the wealth-in-the-utility assumption resolved all the anomalies of the New Keynesian model at the zero lower bound. The paper appeared in the *Review of Economics and Statistics* in 2021 (Michaillat and Saez 2021b). In chapter 14 we use the approach developed in the REStat paper to analyze forward guidance in the slackish business cycle model. We find that thanks to the wealth-in-the-utility assumption, the effects of forward guidance are reasonable, and in fact dissipate as the zero-lower-bound episode lasts an increasingly long time.

3.5. How large should stimulus packages be?

When monetary policy is constrained by the zero lower bound—as during the Great Depression and the Great Recession—the Federal Reserve is not able to stabilize the unemployment rate to its efficient level. In that case, monetary policy is ineffective, but government spending can be used to stabilize unemployment. This is why the US

[5] See for instance Zou (1994), Bakshi and Chen (1996), Carroll (2000), Piketty (2011), and Kumhof, Ranciere, and Winant (2015).

government designed a large stimulus package during the Great Recession. But it was unclear at the time how large that stimulus package should be, or even if a stimulus package was desirable at all, because there was no framework to answer such a question.

The policy debate motivated Emmanuel and I to study optimal stimulus spending when there is unemployment, and in particular unemployment is inefficiently high. Public economics studied the optimal provision of public goods in normal times, but not in bad times (Samuelson 1954, 1955). Macroeconomics focused on the size of fiscal multipliers in bad times, but not on the optimal amount of government spending in bad times (Auerbach and Gorodnichenko 2012, 2013). Using a sufficient-statistics approach again, we established that optimal government spending deviates from the Samuelson rule to reduce, but not eliminate, the unemployment gap. So in a typical slump, in which unemployment is inefficiently high but government spending is able to reduce unemployment, the government should spend above the Samuelson rule—stimulus spending should be positive. The size of the stimulus package depends on three sufficient statistics: the unemployment gap, the fiscal multiplier, and the elasticity of substitution between public and private goods. These results were published in the *Review of Economic Studies* in 2019, and are covered in chapter 18 (Michaillat and Saez 2019).

3.6. Measuring the unemployment gap

In the slackish model, unlike in neoclassical models, efficiency is not guaranteed. Since the amount of slack is generally inefficient, policies might help stabilize the economy. As the AEJ and REStud papers show, optimal policies depend critically on whether the current unemployment rate is above or below the socially efficient unemployment rate. The issue is that we did not know what the efficient unemployment rate, u^*, was.

So our next project was to develop a measure of the socially efficient unemployment rate and of the unemployment gap. We found that the socially efficient unemployment rate depended on three sufficient statistics: the cost of recruiting workers, the social value of nonwork, and the elasticity of the Beveridge curve. When we applied the measure to the United States, we found that the US unemployment gap is generally positive and sharply countercyclical. This means that the US labor market is generally inefficient and especially inefficiently slack in slumps. This implies—going back to the REStud paper—that government spending should be countercyclical. This project was published in the *Journal of Public Economics Plus* in 2021 (Michaillat and Saez 2021a). The results are covered in chapter 10.

In the course of our work on the JPubE paper, Emmanuel and I uncovered that in the United States, servicing a vacancy requires about one worker, home production by jobseekers is virtually nonexistent, and the Beveridge curve is close to a rectangular hyperbola. As a result, our formula for the efficient unemployment rate can be simplified to

$u^* = \sqrt{uv}$. The value u^* also measures the full-employment rate of unemployment (FERU), because legal texts define full employment as the amount of employment maximizing social welfare. We also find that since 1929—so for almost a century, since the Great Depression—the US economy has generally been inefficiently slack. In other words, the US economy has generally operated short of full employment. These results were published in the *Brookings Papers on Economic Activity* in 2024 (Michaillat and Saez 2024b). They are presented in chapter 13.

3.7. Linking inflation to slack

A lot of the work in the book was motivated by the significant slack that appeared in the US economy after the Great Recession and during its long recovery. The recovery of the pandemic recession of 2020, by contrast, was fast and led to significant tightness in the US labor market. In fact the US labor market had not experienced such tight conditions since the end of World War 2. These tight conditions led to new questions.

The first is whether the inefficient tightness experienced on the US labor market could explain the high inflation experienced at the same time, and if so through which mechanism. Until then Emmanuel and I had stuck to rigid inflation rules because inflation had remained exceedingly stable in the thirty years prior to the pandemic. But the inflation spike after the pandemic pushed us to study how slack and inflation could be connected. We propose a new theory of the Phillips curve that directly connects the unemployment gap to inflation, and which predicts that inflation is above its target any time that the economy is inefficiently tight—in line with the evidence unearthed by Benigno and Eggertsson (2023). This work was circulated on arXiv (Michaillat and Saez 2024a) and is described in chapter 15.

3.8. Does immigration affect slack?

A second question that arose during the post-pandemic overheating is whether there were other policies besides cutting aggregate demand that could help. Naturally, another policy that can help stabilize the economy is to expand aggregate supply via immigration. I find that immigration indeed reduces labor market tightness, and that the effect of immigration on tightness and unemployment depends on the state of the labor market. In particular, when the labor market becomes slacker, local workers are more negatively impacted by immigration, as the competition for jobs is fiercer. I also argue that while immigration has a negative effect on local workers because it increases their unemployment rate, it has a positive effect on local firms because it makes it easier for them to recruit—explaining why immigration is such a polarizing topic. This work on migration was circulated on arXiv (Michaillat 2024) and is discussed in chapter 12.

3.9. Has a new recession started?

Finally, after labor market tightness peaked in 2022, the US labor market cooled fairly rapidly in 2023–2025. This naturally led to the question: Has a new recession started? The JPubE and BPEA papers show that the combination of vacancy and unemployment data is useful to compute the FERU and unemployment gap in real time. But it turns out that these data are very useful to detect recessions in real time too. Unemployment data, combined with simple threshold rules, provide a more reliable signal of US recessions than other detection methods—especially those based on the gross domestic product (Crump, Giannone, and Lucca 2020a,b). The logic behind unemployment-threshold rules is simple: unemployment always goes up in recessions, so a recession can be detected when the unemployment rate increases sharply.

However, the unemployment rate is only a noisy measure of the latent state of the labor market. Another measure is the vacancy rate, as recessions feature not only an increase in the unemployment rate but also a decline in the vacancy rate as the economy moves along the Beveridge curve. Therefore, by combining data on unemployment and job vacancies, we obtain a recession indicator that is less noisy than unemployment-based indicators. Thanks to the reduced noisiness, the detection threshold can be lowered once both unemployment and vacancy data are used. The rule that we construct (Michez rule) detects recessions faster than the popular Sahm (2019) rule: with an average delay of 1.2 months instead of 2.7 months, and a maximum delay of 3 months instead of 7 months. It is also more robust: it identifies the 15 recessions that occurred since 1929 without false positives, while the Sahm rule fails before 1960. This work was published in the *Oxford Bulletin of Economics and Statistics* in 2025 (Michaillat and Saez 2025).

While the Sahm rule, Michez rule, and other threshold rules choose their thresholds optimally, they filter the data in an arbitrary way. As such, they may not extract the most information possible from unemployment and vacancy data. To address this issue, I developed a method to optimize how recession classifiers are constructed—to extract the most information possible from labor market data. By optimally filtering the data and selecting the threshold, recessions can be detected even more rapidly and accurately than with the Michez or Sahm rule. This method identifies a collection of recession rules along an anticipation-precision frontier, and in chapter 19, I present one of these optimal rules. The rule presented here (Michez+ rule) is optimal—in that it lies on the anticipation-precision frontier—and is simple to construct. This work on the anticipation-precision frontier was circulated on arXiv (Michaillat 2025).

3.10. So what is new in this book?

Readers familiar with the research papers described in this chapter will recognize some of the elements of the slackish model and policy analysis discussed in the book. The book is not a simple collection of papers, however: it provides a more comprehensive treatment of the material than is possible in individual papers.

First, the book offers a unified and systematic statement of the theory. Whereas individual papers necessarily focus on specific questions in isolation, the book integrates the research into a coherent theoretical framework. All the models and policies are presented and analyzed with consistent notation and methodology.

Second, the book provides a more extensive empirical grounding. It aggregates evidence from numerous sources and time periods for the United States. This allows for a comprehensive empirical assessment of the slackish framework that spans nearly a century of economic data.

Third, the book relates the slackish framework to the main business cycle paradigms. It draws comparisons to the Old Keynesian, Real Business Cycle, and New Keynesian models to delineate where the assumptions and predictions overlap and where they differ. This clarifies how the slackish model provides a different lens through which we can look at business cycle fluctuations and rethink the design of stabilization policies.

Bibliography

Akerlof, George A. 1984. "Gift Exchange and Efficiency-Wage Theory: Four Views." *American Economic Review* 74 (2): 79–83. `http://www.jstor.org/stable/1816334`.

Akerlof, George A. and Janet L. Yellen. 1990. "The Fair Wage-Effort Hypothesis and Unemployment." *Quarterly Journal of Economics* 105 (2): 255–283. `https://doi.org/10.2307/2937787`.

Albrecht, James. 2011. "Search Theory: The 2010 Nobel Memorial Prize in Economic Sciences." *Scandinavian Journal of Economics* 113 (2): 237–259. `https://doi.org/10.1111/j.1467-9442.2011.01658.x`.

Asher, Martin A., Robert H. Defina, and Kishor Thanawala. 1993. "The Misery Index: Only Part of the Story." *Challenge* 36 (2): 58–62. `https://doi.org/10.1080/05775132.1993.11471656`.

Auerbach, Alan J. and Yuriy Gorodnichenko. 2012. "Measuring the Output Responses to Fiscal Policy." *American Economic Journal: Economic Policy* 4 (2): 1–27. `https://doi.org/10.1257/pol.4.2.1`.

Auerbach, Alan J. and Yuriy Gorodnichenko. 2013. "Fiscal Multipliers in Recession and Expansion." In *Fiscal Policy after the Financial Crisis*, edited by Alberto Alesina and Francesco Giavazzi, chap. 2. Chicago: University of Chicago Press. `https://www.nber.org/chapters/c12634`.

Baily, Martin N. 1978. "Some Aspects of Optimal Unemployment Insurance." *Journal of Public Economics* 10 (3): 379–402. `https://doi.org/10.1016/0047-2727(78)90053-1`.

Bakshi, Gurdip S. and Zhiwu Chen. 1996. "The Spirit of Capitalism and Stock-Market Prices." *American Economic Review* 86 (1): 133–157. `https://www.jstor.org/stable/2118259`.

Barro, Robert J. and Herschel I. Grossman. 1971. "A General Disequilibrium Model of Income and Employment." *American Economic Review* 61 (1): 82–93. `http://www.jstor.org/stable/1910543`.

Benigno, Pierpaolo and Gauti B. Eggertsson. 2023. "It's Baaack: The Surge in Inflation in the 2020s and the Return of the Non-Linear Phillips Curve." NBER Working Paper 31197. `https://doi.org/10.3386/w31197`.

Blanchard, Olivier J. and Jordi Gali. 2010. "Labor Markets and Monetary Policy: A New Keynesian Model with Unemployment." *American Economic Journal: Macroeconomics* 2 (2): 1–30. `https://doi.org/10.1257/mac.2.2.1`.

Carroll, Christopher D. 2000. "Why Do the Rich Save So Much?" In *Does Atlas Shrug? The Economic Consequences of Taxing the Rich,* edited by Joel B. Slemrod. Cambridge, MA: Harvard University Press.

Chetty, Raj. 2006. "A General Formula for the Optimal Level of Social Insurance." *Journal of Public Economics* 90 (10-11): 1879–1901. https://doi.org/10.1016/j.jpubeco.2006.01.004.

Chetty, Raj. 2009. "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced-Form Methods." *Annual Review of Economics* 1: 451–488. https://doi.org/10.1146/annurev.economics.050708.142910.

Crump, Richard, Domenico Giannone, and David Lucca. 2020a. "Reading the Tea Leaves of the US Business Cycle – Part One." Liberty Street Economics, Federal Reserve Bank of New York. https://perma.cc/W79A-EFPF.

Crump, Richard, Domenico Giannone, and David Lucca. 2020b. "Reading the Tea Leaves of the US Business Cycle – Part Two." Liberty Street Economics, Federal Reserve Bank of New York. https://perma.cc/8CEL-TK37.

Diamond, Peter. 2011. "Unemployment, Vacancies, Wages." *American Economic Review* 101 (4): 1045–1072. https://doi.org/10.1257/aer.101.4.1045.

Eyster, Erik, Kristof Madarasz, and Pascal Michaillat. 2021. "Pricing under Fairness Concerns." *Journal of the European Economic Association* 19 (3): 1853–1898. https://doi.org/10.1093/jeea/jvaa041.

Hall, Robert E. 2005. "Employment Fluctuations with Equilibrium Wage Stickiness." *American Economic Review* 95 (1): 50–65. https://doi.org/10.1257/0002828053828482.

Katz, Lawrence F. 1986. "Efficiency Wage Theories: A Partial Evaluation." *NBER Macroeconomics Annual* 1: 235–276. https://doi.org/10.1086/654025.

Keynes, John Maynard. 1936. *The General Theory of Employment, Interest and Money*. New York: Macmillan.

Kleven, Henrik J. 2021. "Sufficient Statistics Revisited." *Annual Review of Economics* 13: 515–538. https://doi.org/10.1146/annurev-economics-060220-023547.

Kumhof, Michael, Romain Ranciere, and Pablo Winant. 2015. "Inequality, Leverage, and Crises." *American Economic Review* 105 (3): 1217–1245. https://doi.org/10.1257/aer.20110683.

Kurz, Mordecai. 1968. "Optimal Economic Growth and Wealth Effects." *International Economic Review* 9 (3): 348–357. https://doi.org/10.2307/2556231.

Landais, Camille, Pascal Michaillat, and Emmanuel Saez. 2018a. "A Macroeconomic Approach to Optimal Unemployment Insurance: Applications." *American Economic Journal: Economic Policy* 10 (2): 182–216. https://doi.org/10.1257/pol.20160462.

Landais, Camille, Pascal Michaillat, and Emmanuel Saez. 2018b. "A Macroeconomic Approach to Optimal Unemployment Insurance: Theory." *American Economic Journal: Economic Policy* 10 (2): 152–181. https://doi.org/10.1257/pol.20150088.

Malinvaud, Edmond. 1977. *The Theory of Unemployment Reconsidered*. Oxford: Basil Blackwell.

Michaillat, Pascal. 2012. "Do Matching Frictions Explain Unemployment? Not in Bad Times." *American Economic Review* 102 (4): 1721–1750. https://doi.org/10.1257/aer.102.4.1721.

Michaillat, Pascal. 2014. "A Theory of Countercyclical Government Multiplier." *American Economic Journal: Macroeconomics* 6 (1): 190–217. https://doi.org/10.1257/mac.6.1.190.

Michaillat, Pascal. 2024. "Modeling Migration-Induced Unemployment." arXiv:2303.13319. https://doi.org/10.48550/arXiv.2303.13319.

Michaillat, Pascal. 2025. "Recession Detection Using Classifiers on the Anticipation-Precision Frontier." arXiv:2506.09664. https://doi.org/10.48550/arXiv.2506.09664.

Michaillat, Pascal and Emmanuel Saez. 2015. "Aggregate Demand, Idle Time, and Unemployment." *Quarterly Journal of Economics* 130 (2): 507–569. https://doi.org/10.1093/qje/qjv006.

Michaillat, Pascal and Emmanuel Saez. 2019. "Optimal Public Expenditure with Inefficient Unemployment." *Review of Economic Studies* 86 (3): 1301–1331. https://doi.org/10.1093/restud/rdy030.

Michaillat, Pascal and Emmanuel Saez. 2021a. "Beveridgean Unemployment Gap." *Journal of Public Economics Plus* 2: 100009. https://doi.org/10.1016/j.pubecp.2021.100009.

Michaillat, Pascal and Emmanuel Saez. 2021b. "Resolving New Keynesian Anomalies with Wealth in the Utility Function." *Review of Economics and Statistics* 103 (2): 197–215. https://doi.org/10.1162/rest_a_00893.

Michaillat, Pascal and Emmanuel Saez. 2022. "An Economical Business-Cycle Model." *Oxford Economic Papers* 74 (2): 382–411. https://doi.org/10.1093/oep/gpab021.

Michaillat, Pascal and Emmanuel Saez. 2024a. "Beveridgean Phillips Curve." arXiv:2401.12475. https://doi.org/10.48550/arXiv.2401.12475.

Michaillat, Pascal and Emmanuel Saez. 2024b. "$u^* = \sqrt{uv}$: The Full-Employment Rate of Unemployment in the United States." *Brookings Papers on Economic Activity* 55 (2): 323–390. https://doi.org/10.1353/eca.2024.a964373.

Michaillat, Pascal and Emmanuel Saez. 2025. "Has the Recession Started?" *Oxford Bulletin of Economics and Statistics* 87 (6): 1047–1058. https://doi.org/10.1111/obes.12685.

Mortensen, Dale T. 2011. "Markets with Search Friction and the DMP Model." *American Economic Review* 101 (4): 1073–1091. https://doi.org/10.1257/aer.101.4.1073.

Okun, Arthur M. 1981. *Prices and Quantities: A Macroeconomic Analysis*. Washington, DC: Brookings Institution.

Piketty, Thomas. 2011. "On the Long-Run Evolution of Inheritance: France 1820–2050." *Quarterly Journal of Economics* 126 (3): 1071–1131. https://doi.org/10.1093/qje/qjr020.

Pissarides, Christopher A. 2000. *Equilibrium Unemployment Theory*. 2nd ed. Cambridge, MA: MIT Press.

Pissarides, Christopher A. 2011. "Equilibrium in the Labor Market with Search Frictions." *American Economic Review* 101 (4): 1092–1105. https://doi.org/10.1257/aer.101.4.1092.

Saez, Emmanuel. 2001. "Using Elasticities to Derive Optimal Income Tax Rates." *Review of Economic Studies* 68 (1): 205–229. https://doi.org/10.1111/1467-937X.00166.

Sahm, Claudia. 2019. "Direct Stimulus Payments to Individuals." In *Recession Ready: Fiscal Policies to Stabilize the American Economy*, edited by Heather Boushey, Ryan Nunn, and Jay Shambaugh, chap. 3. Washington: Brookings Institution. https://www.brookings.edu/articles/direct-stimulus-payments-to-individuals/.

Samuelson, Paul A. 1954. "The Pure Theory of Public Expenditure." *Review of Economics and Statistics* 36 (4): 387–389. https://doi.org/10.2307/1925895.

Samuelson, Paul A. 1955. "Diagrammatic Exposition of a Theory of Public Expenditure." *Review of Economics and Statistics* 37 (4): 350–356. https://doi.org/10.2307/1925849.

Yellen, Janet L. 1984. "Efficiency Wage Models of Unemployment." *American Economic Review* 74 (2): 200–205. https://www.jstor.org/stable/1816355.

Zou, Heng-fu. 1994. "The Spirit of Capitalism and Long-Run Growth." *European Journal of Political Economy* 10: 279–293. https://doi.org/10.1016/0176-2680(94)90020-5.